

PSP - Database Cost Optimization

- Purpose
- Context
- Cost
- Production
- Pre-Production
- Goals
- Principles
- Cost Optimisation Opportunities
- Execution Plan

Purpose

The purpose of this document is to outline a plan for lowering expenses associated with the Payroll platform and services databases, DMS, EC2. Currently, the monthly costs are roughly around **\$316K** as of Nov-2023. We have already reduced cost by **~38% (121K)** in the month of **Jan-2024 (195K)**. There are further opportunities to optimize and reduce cost further by ~50%.

Context

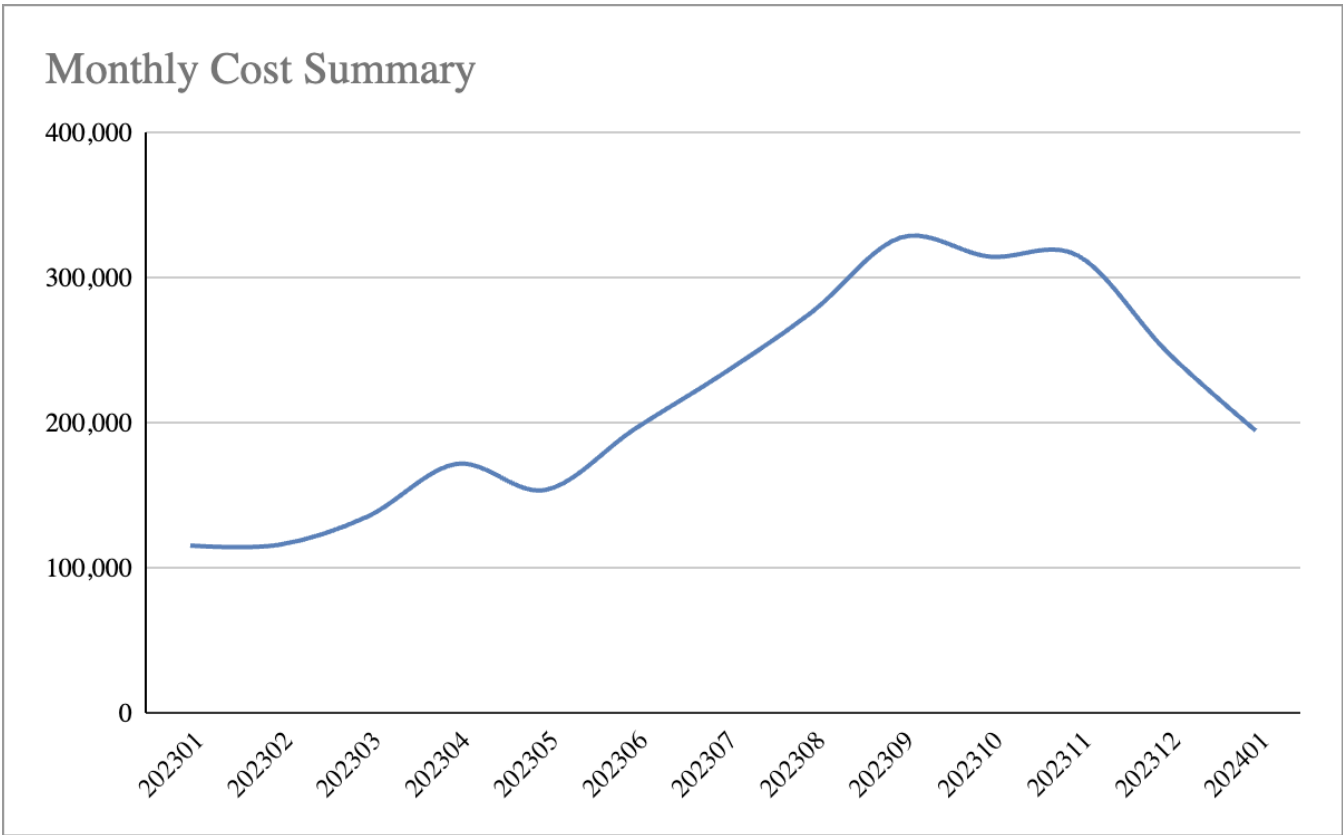
PSP uses two databases i.e. Monolith and audit database, which are migrated to Aurora Postgres. Reader instances are in use by downstream and OLAP jobs.

Cost

Production

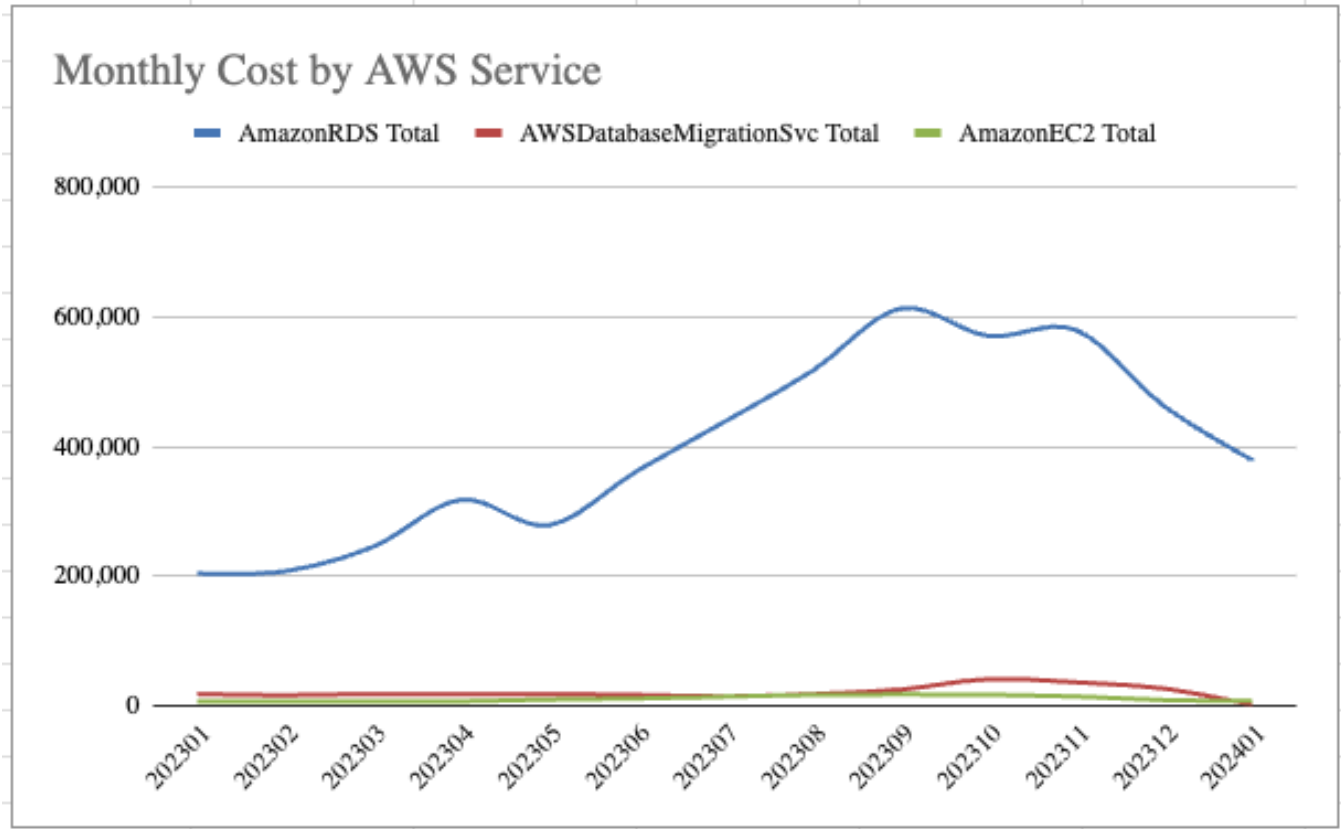
Overall Production Monthly Cost

	202307	202308	202309	202310	202311	202312	202401
1	234,231	276,646	328,092	314,956	315,585	249,316	194,931



Production Monthly Cost at Service level

		202307	202308	202309	202310	202311	202312	202401
1	RDS	219,069	258,278	306,061	284,977	289,463.77	230,848	189,468
2	DMS	7,742	9,445	12,899	20,998	18,482	13,646	1210
3	EC2	7,421	8,922	9,132	8,981	7,639	4,822	4253
4	Total Cost	234,231	276,646	328,092	314,956	315,585	249,316	194,931

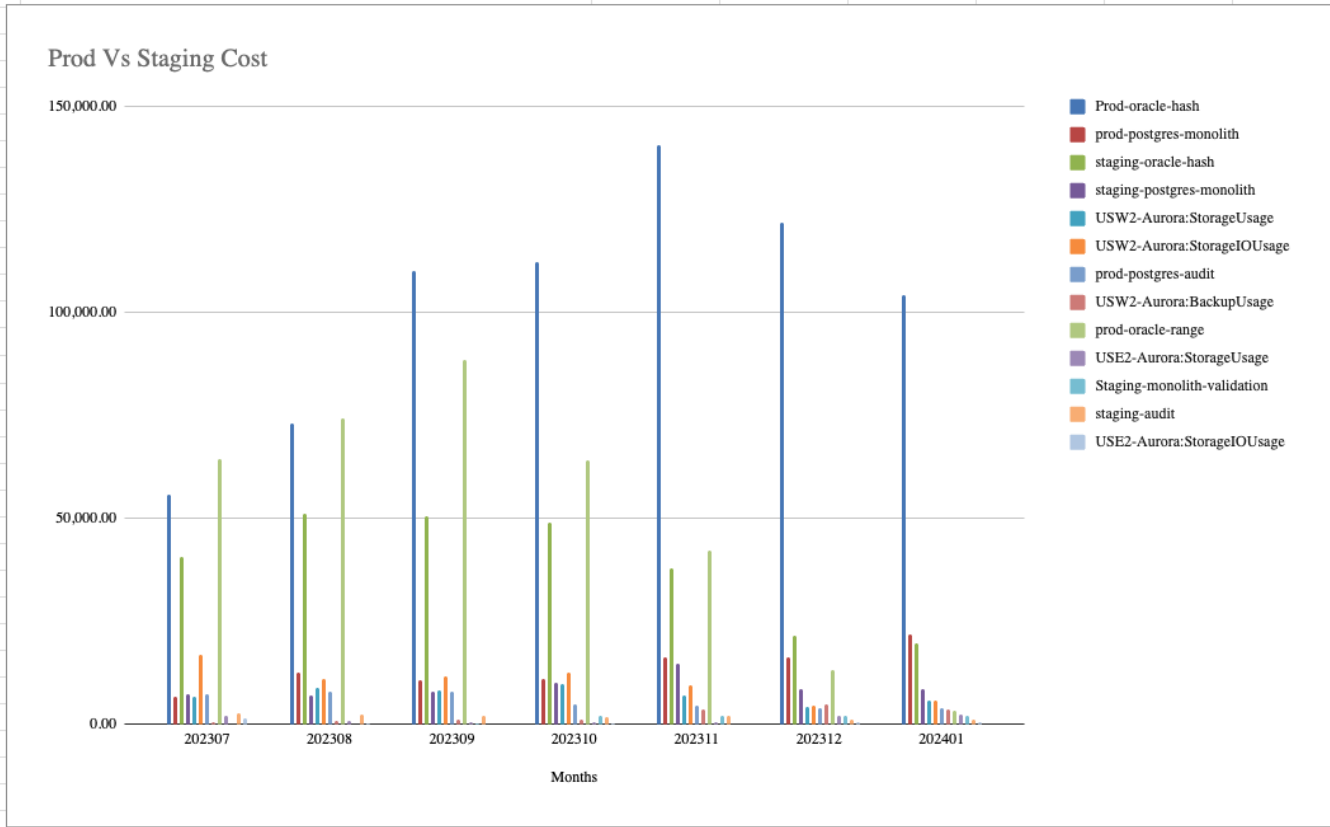


Staging Cost

- Monthly cost is ~30K (including oracle hash). After Oracle decommission, it would be ~10K

Production Cost

- After oracle decommission, cost should come down at ~110K

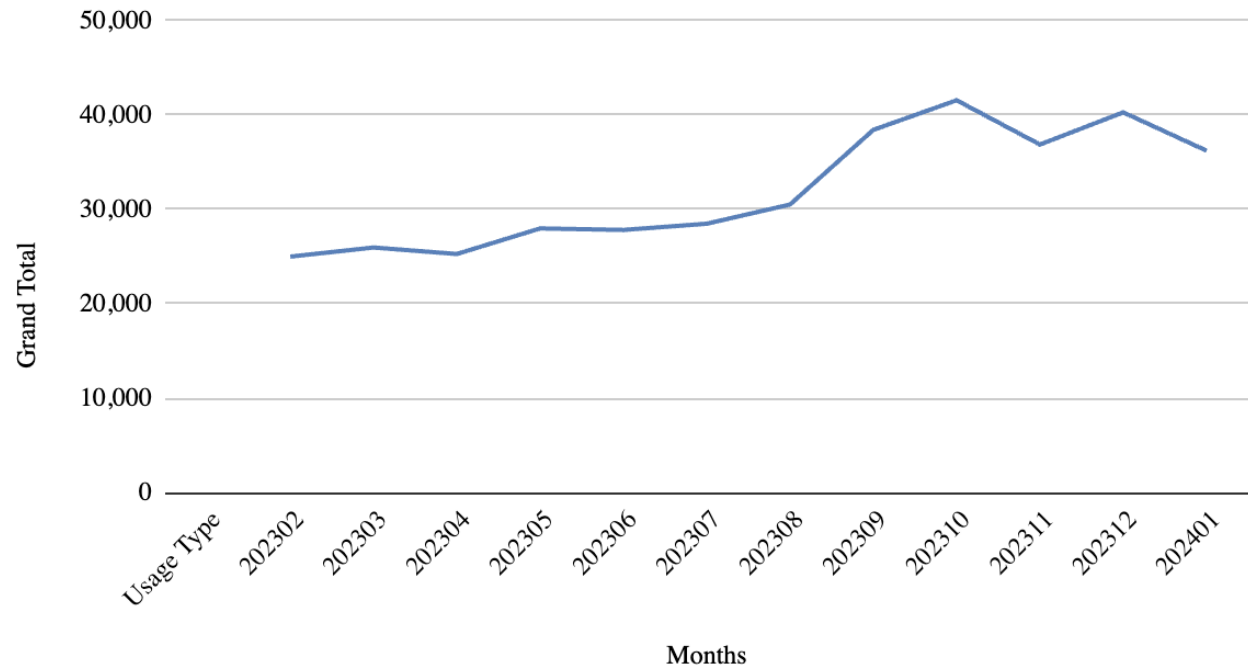


Pre-Production

Overall PreProduction Monthly Cost

202307	202308	202309	202310	202311	202312
29,344	31,327	39,386	42,576	37,738	41,164

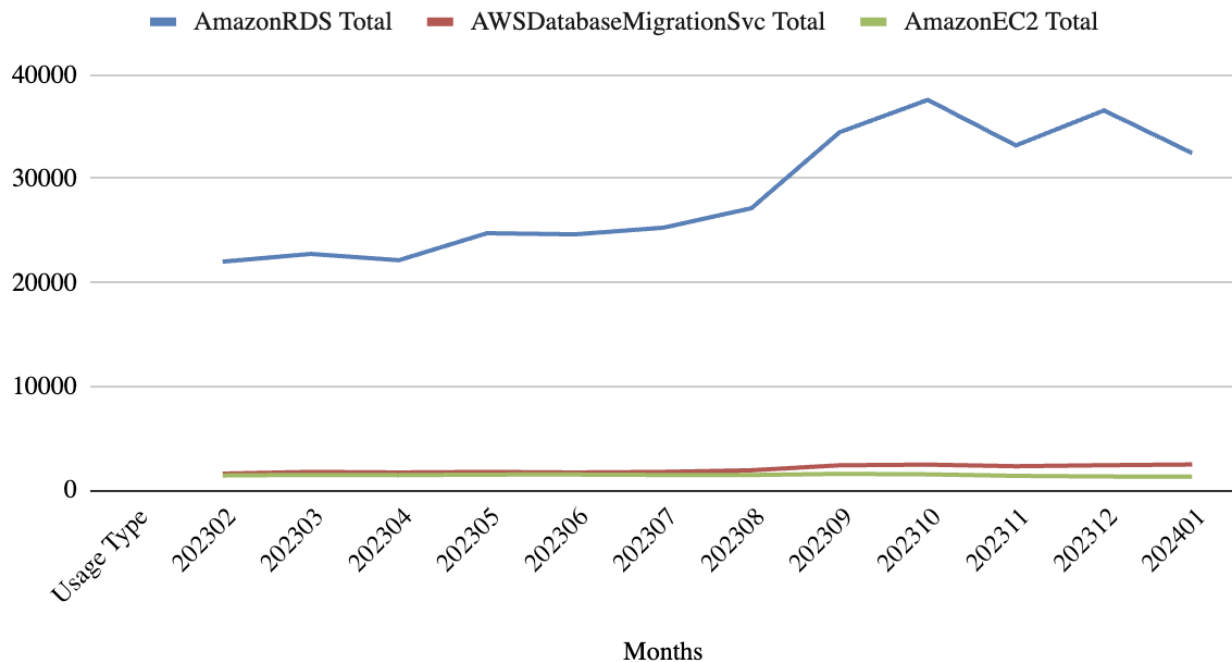
PreProd Monthly Cost



PreProduction Monthly Cost at Service level

	202307	202308	202309	202310	202311	202312	202401
AmazonRDS Total	25,281	27,159	34,479	37,599	33,207	36,579	32,469
AWSDatabaseMigrationSvc Total	1,746	1,905	2,387	2,449	2,297	2,394	2,465
AmazonEC2 Total	1,456	1,453	1,561	1,514	1,368	1,308	1,287
Total	28,483	30,517	38,427	41,562	36,873	40,280	36,221

Monthly Cost by Service



Reference : [PSP - Database Cost Analysis](#)

Goals

Principles

- Whenever feasible, utilize a shared database for non-production environments.
- Make sure that the GG Hub configuration is appropriately sized and utilized until replication is set up. Once the replication cleanup is done, eliminate the GG hubs for replication and data validation.
- Use a Data Lake for OLAP operations like finance reporting, datalake jobs, RJF, Tableau, Accounting) to reduce traffic on reader instances
- Shut down databases when not in use (DS2, PRF, STG)

Cost Optimisation Opportunities

- On demand staging db setup
 - After Oracle decommission, it would save ~10K
- Data archival & purging
- Explore taking CLOB out of audit database
- Review database backup retention period for databases
- Keep lesser instance size of DR cluster instances
- keep less backup retention period for DR clusters
 - Since we enabled monolith DR in Dec-2023 in addition to audit db. Total cost for backup for the month of Jan-2024 is ~500
- Decommission preprod oracle databases
 - savings would be ~15K.
- Have shared cluster for preprod databases without impacting developer productivity.
- Test r6g.16xlarge instance size for production instance

Execution Plan

Category	Tasks	Status	Start Date	Completion Date	ETA	Cost Savings
----------	-------	--------	------------	-----------------	-----	--------------

DMS Replication	Deleted replication from oracle hash(B) to Aurora Postgres(C3) Deleted replication from Aurora Postgres (C3) to Oracle Hash (D3) Deleted 5 Replication instance	Completed	03 Nov 2023	08 Dec 2023		Saved ~15K
Goldengate Replication	Deleted replication from Oracle Hash (B) to Oracle range (A) and reverse Deleted replication from Oracle Range (A) to Oracle range DR (P) Deleted 2 GG hubs in us-east Deleted 1 GG hub in us-west	Completed	03 Nov 2023	17 Nov 2023		Saved ~5K
	Remove replication from Aurora Postgres to Oracle Hash Delete 2 gg hubs for reverse replication	Not started	01 Mar 2024		31 Mar 2024	
	Remove Veridata server and repository database	Not started	01 Mar 2024		31 Mar 2024	Savings ~3K
Databases	Dropped Oracle Range DR in us-east-2 (pspuep05) Dropped Oracle Range in us-west-2 (pspuwp01 & its read replica pspadg01) Dropped Oracle Hash psphp06 (D3) Dropped Aurora Postgres psp-prod-mon (C3) Dropped Oracle Hash database in us-east-2 (Q)	Completed	03 Nov 2023	17 Nov 2023		Saved ~100K
	Drop Oracle hash databases (B)	Not started	21 Feb 2024		29 Feb 2024	Savings ~70K
	Drop Oracle hash databases (D2)	In Progress	21 Feb 2024		29 Feb 2024	Savings ~25K
	Reduce production db instance size	Not started				
	Drop 2 reader instances in us-east-2(DR) region	Not started	21 Feb 2024		29 Feb 2024	Savings ~3.8K
	Audit DB removal	In Progress	21 Feb 2024		31 May 2024	Savings ~5K
	Drop Oracle Staging databases Drop Staging postgres validation database (psp-stg-pspval)	Completed	01 Feb 2024	18 Feb 2024		Savings ~22K
Backups	Cleanup manual backups	In Progress	21 Feb 2024		29 Feb 2024	
	Disable backup of staging databases	In Progress	20 Feb 2024		29 Feb 2024	
Data Purging & Archival	Archive & Purge data	Not started				
Preprod	Review backup plans for databases	In Progress	20 Feb 2024		31 Mar 2024	Savings ~10K
	Drop Oracle databases, Veridata	In Progress	20 Feb 2024		31 Mar 2024	Savings ~5K
	Plan & Implement non-prod env in shared mode	Not started				